Speech-Based Stress Detection with Handcrafted and Self-Supervised Features

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Motivation

Why Stress?

Stress is a physiological and psychological response, triggered by factors like work, cognitive load, and anxiety. It can be categorized into acute (short-term) and chronic (long-term) stress, both of which impact productivity, mental health, and sleep. Stress-related disorders are on the rise globally, highlighting the need for efficient, non-invasive detection methods

Why Speech?

Speech is convenient to capture and contains prosodic markers (e.g., pitch, rate) that change with stress. These markers correlate with physiological stress responses (e.g., cortisol) and can help identify stress levels (Kappen et al., 2022).

Objective

Compare handcrafted and self-supervised feature extraction approaches for detecting acute stress from speech.

Explore an ensemble system that combines both methods for improved performance.

Aim to develop a real-time stress detection system for applications in workplaces (e.g., air traffic control) and mental health assessments.

Datasets

1. StressID Dataset:

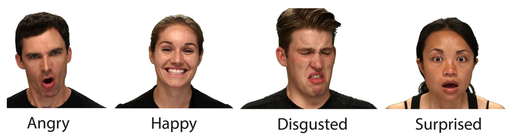
- Includes 65 participants performing 11 tasks (e.g., emotional video clips, cognitive tasks, public speaking) to induce varied stress responses, self-ratings of relaxation, stress, arousal, and valence are provided for each task.

Task	Baseline	Counting1	Counting2	Counting3	Math	Reading	Relax	Speaking	Stroop	Video1	Video2
Included?	✘	✔	✔	✔	✔	✔	✘	✔	✔	✘	✘

- **Labeling Schemes for StressID:**
  a. Binary Stress Classification: Stress thresholded at 5 on a 0-10 scale (≥5 = stressed, <5 = not stressed).
  b. Three-Class Affect-Based Labeling: Labels derived from valence, arousal, and relaxation/stress ratings.
  c. Stress Regression: Self-reported 0-10 scale used as a continuous target.

2. RAVDESS Dataset:

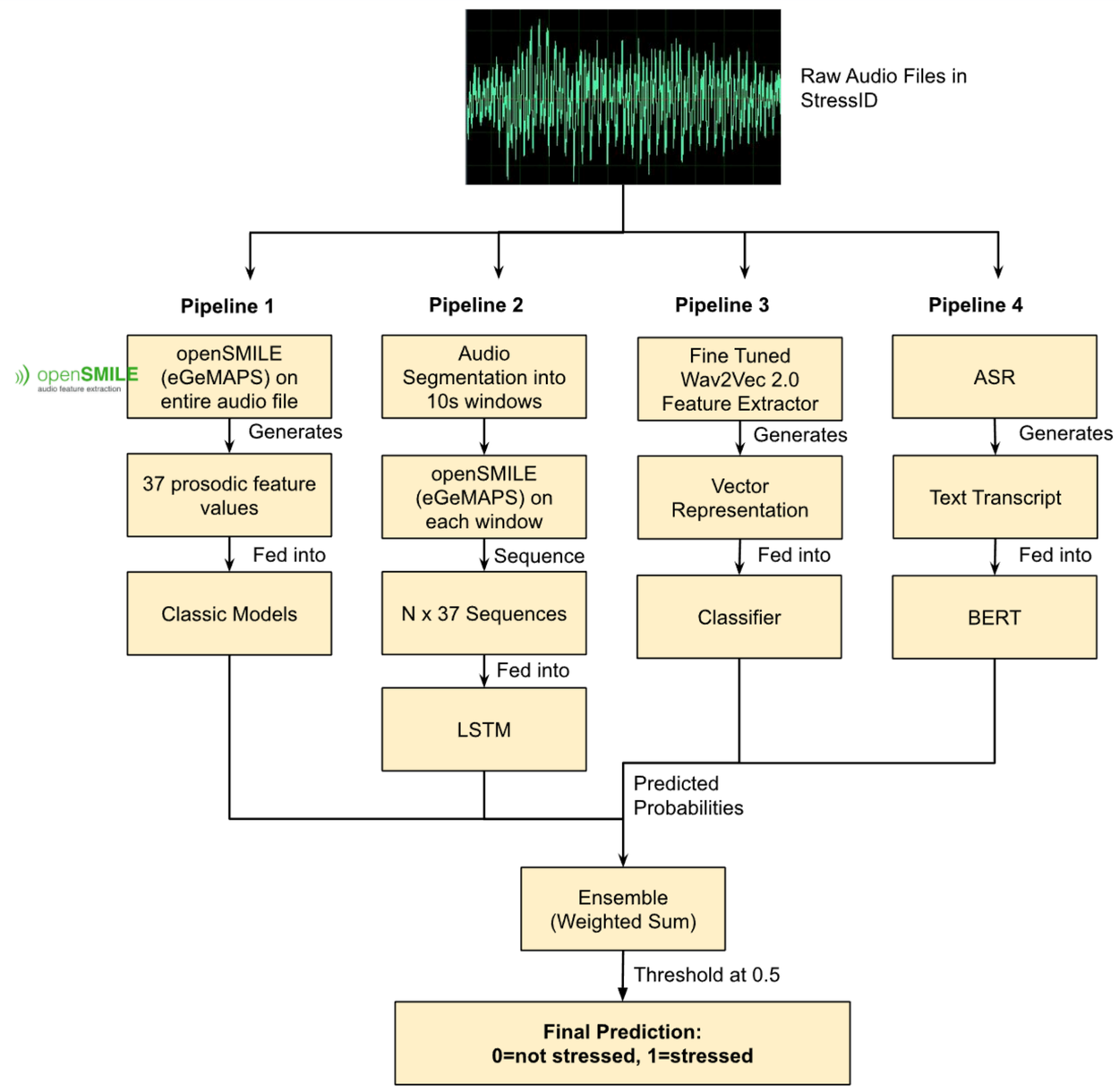
- Contains 24 professional actors expressing 8 emotions: calm, happy, sad, angry, fearful, surprise, disgust, and neutral.
- Used for pretraining emotion recognition models to learn prosodic and affective cues for the Wav2Vec 2.0 feature extractor.



Train-Test Split:

- 80/20 speaker-independent split, ensuring no overlap of speakers between training and testing sets.
- Class Imbalance: StressID is imbalanced (approximately 2/3 stressed, 1/3 not stressed). Random upsampling was applied to the minority class in the training set.
- Benchmark: Balanced accuracy (Bal Acc.) was used as the main evaluation metric on the validation set.

Proposed Methods



Results and Discussion

Pipelines	Representation	Model	Bal Acc	F1 (macro)	AUC (ROC-AUC)	Precision (stressed)	Recall (stressed)
Baseline (majority class)	-	-	0.50	0.37	0.50	0.59	1.00
openSMILE (eGeMAPS) → classical ML	eGeMAPSv02 (37d)	SVM (RBF, tuned)	0.64	0.64	0.50	0.64	0.64
		GB (Tuned)	0.60	0.58	0.62	0.69	0.60
		SVC (Baseline)	0.52	0.50	0.62	0.52	0.52
		Logistic Reg.	**0.69**	**0.69**	**0.76**	**0.69**	**0.69**
		Random Forest	0.59	0.55	0.55	0.75	0.59
openSMILE (windowed) → LSTM	Windowed eGeMAPS (37d) sequence	LSTM	0.66	0.66	0.73	0.69	0.83
Wav2Vec2 fine-tuned classifier	Wav2Vec2 embeddings (finetuned)	Wav2Vec2 classifier	0.64	0.63	0.67	0.67	0.93
ASR → BERT text classifier	ASR transcript	BERT-base classifier	0.44	0.35	0.55	1.00	0.05
Ensemble (weighted)	**Probabilities from each pipeline**	**Weighted mean**	**0.70**	**0.70**	**0.81**	**0.74**	**0.78**

- Overall, almost all pipelines beat the StressID baseline, except ASR→BERT (text), due to StressID speech having low lexical content.
- Logistic Regression was the strongest classical approach, reaching 69% balanced accuracy.
- Notably, deep models underperformed. Wav2Vec2.0 and openSMILE(windowed) → LSTM did worse than classic ML; their high recall + low precision suggests they predict "stress" too often even after attempts to upsample to balance classes.
- There were too little labelled StressID data to fine-tune large models → overfitting and unstable thresholds; windowed LSTM may overemphasize short prosodic fluctuations, biasing toward "stressed."
- Best overall approach: The ensemble performed best (70% balanced accuracy, ROC-AUC 0.81), improving on the 62% StressID baseline, by combining robust handcrafted-feature models with additional deep-representation signals for better stability across users/tasks.

Conclusion

- Prosody works best: Handcrafted pitch/energy/voice-quality features gave the top results; self-supervised features were similar but not reliably better—limits likely from small/noisy data, not model capacity.
- StressID limits: Only ~6 hours of speech and subjective self-rated labels, so deep-model fine-tuning (e.g., Wav2Vec2.0) can overfit and produce unstable thresholds in speaker-independent tests.
- Next steps: Add physiological signals (ECG/EDA/RR) to score stress relative to each person's baseline and validate/improve labels by checking alignment with self-reports.
- Code: https://github.com/StifflenTwee/RO047

References

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